PADHAKU
Science · Class 10 Science



CHAPTER 7

How do Organisms Reproduce?

NCERT Exercise Solutions

Questions · 7.1 Do Organisms Create Exact Copies of Themselves?	2 Q
Activities · 7.2 Modes of Reproduction Used by Single Organisms	6 Q
Questions · 7.2 Modes of Reproduction Used by Single Organisms	5 Q
Activity · 7.3 Sexual Reproduction	1 Q
Questions · 7.3 Sexual Reproduction	5 Q
Exercises	11 Q

✨ Try each question yourself first, then check the answer. 💡 answers are samples: write your own.

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Questions · 7.1 Do Organisms Create Exact Copies of Themselves?

Q 1 **What is the importance of DNA copying in reproduction?**

✔ ANSWER

- DNA in the nucleus carries the information for making proteins, which decide the body design.
- DNA copying passes this blueprint to the new individual, so offspring look like the parents.
- Small errors in copying create **variations**, which help the species survive and are the basis of evolution.

Q 2 **Why is variation beneficial to the species but not necessarily for the individual?**

✔ ANSWER

- If the niche changes (e.g. water gets hotter), most individuals may die, but a few variants (e.g. heat-resistant bacteria) survive and keep the species going.
- So variation helps the **species** survive over time.
- For one individual, a variation may be useless or even harmful (a drastic change can make the cell die), so it is not necessarily beneficial to the individual.

Activities · 7.2 Modes of Reproduction Used by Single Organisms

Activity 7.1 **Put a drop of yeast grown in sugar solution (1–2 hours, warm place) on a slide and observe under a microscope.**

✔ ANSWER

Observation: Yeast cells show small bulb-like outgrowths (buds); some buds form chains of cells.

Conclusion: Yeast reproduces asexually by **budding**.

Activity 7.2 **Keep a wet slice of bread in a cool, moist, dark place and observe it with a magnifying glass for a week. Compare and contrast the ways in which yeast grows in the first case, and how mould grows in the second.**

✔ ANSWER

Observation: White cotton-like threads (hyphae) grow on the bread; later tiny black blob-on-a-stick structures (sporangia) appear.

- Yeast (unicellular) grows by **budding**: a small bud grows from the parent cell and separates.
- Bread mould (Rhizopus, multicellular) spreads by thread-like hyphae and reproduces by **spores** formed in sporangia.

- Both are asexual and need moisture and food (sugar / bread).

Activity 7.3 **Observe permanent slides of Amoeba and of Amoeba showing binary fission, and compare them.**

✔ ANSWER

Observation: The first slide shows a single Amoeba with one nucleus. The second shows the nucleus divided into two and the cytoplasm splitting, forming two daughter Amoebae.

Conclusion: Amoeba reproduces by **binary fission**; the split can take place in any plane.

Activity 7.4 **Observe filaments of Spirogyra (from dark green pond water) under a microscope. Can you identify different tissues in the Spirogyra filaments?**

✔ ANSWER

Observation: The filament is a chain of similar cells, each with a ribbon-like spiral chloroplast and a nucleus.

No, different tissues cannot be seen; all cells are alike.

Conclusion: Since every cell is similar and can do all functions, any fragment can grow into a new Spirogyra. This is why it can reproduce by **fragmentation**.

Activity 7.5 **Cut a potato into pieces, some with a notch (bud/eye) and some without, and keep them on wet cotton. Which are the potato pieces that give rise to fresh green shoots and roots?**

✔ ANSWER

Yes, notches (eyes) can be seen on the potato surface.

Observation: Only the pieces **with a notch (bud)** give rise to green shoots and roots; pieces without buds do not grow.

Conclusion: Buds in the notches of the potato tuber (a stem) can develop into new plants: **vegetative propagation**.

Activity 7.6 **Cut money-plant pieces, some with at least one leaf and some from between two leaves, and dip them in water. Which ones grow and give rise to fresh leaves? What can you conclude?**

✔ ANSWER

Observation: The pieces containing a **leaf (with a node)** grow roots and fresh leaves; the pieces cut between two leaves (internode only) do not grow.

Conclusion: New plants arise from buds at the **nodes** (where leaves join the stem). The money-plant can be grown by vegetative propagation from stem cuttings having a node.

? Questions · 7.2 Modes of Reproduction Used by Single Organisms

Q 1 How does binary fission differ from multiple fission?

✔ ANSWER

Binary fission	Multiple fission
The parent cell divides into **two** daughter cells.	The parent cell divides into **many** daughter cells at the same time.
e.g. Amoeba, Leishmania, bacteria	e.g. Plasmodium (malarial parasite)
Usually in favourable conditions.	Often inside a protective cyst in unfavourable conditions.

Q 2 How will an organism be benefited if it reproduces through spores?

✔ ANSWER

- A sporangium makes a **large number** of spores, so many new individuals form.
- Spores have **thick walls** that protect them in unfavourable conditions until they reach a moist surface.

- Spores are light and spread easily by air to new places.

Q 3 **Can you think of reasons why more complex organisms cannot give rise to new individuals through regeneration?**

✔ ANSWER

- Complex organisms have specialised cells organised into tissues, organs and organ systems placed at definite positions.
- A cut piece does not have all the specialised cells and the organised development needed to rebuild the whole body.
- Their body parts are interdependent; a small piece cannot carry out all life processes, so regeneration of a whole organism is not possible.

Q 4 **Why is vegetative propagation practised for growing some types of plants?**

✔ ANSWER

- Plants bear flowers and fruits **earlier** than plants grown from seeds.
- Plants that have lost the capacity to produce seeds (banana, orange, rose, jasmine) can be grown.
- All new plants are genetically similar to the parent, so good characters are kept.

- Many plants (e.g. sugarcane, roses, grapes) can be grown quickly by layering or grafting.

Q 5 **Why is DNA copying an essential part of the process of reproduction?**

✔ ANSWER

- DNA carries the information (blueprint) for making proteins and hence the body design.
- When a cell reproduces, it must make a copy of its DNA so that each new cell gets the full set of information to carry out life processes.
- This keeps the body design of the species the same; small copying errors bring useful variations.

🧪 Activity · 7.3 Sexual Reproduction

Activity 7.7 **Soak Bengal gram (chana) seeds overnight, keep them under a wet cloth for a day, cut them open and identify the parts (compare with Fig. 7.9).**

✔ ANSWER

Observation: The seed has a seed coat; inside are two **cotyledons** (food store) and the embryo with a **plumule** (future shoot) and a **radicle** (future root).

Conclusion: The seed contains the future plant (embryo) and stored food, which help it germinate into a seedling under suitable conditions.

❓ Questions · 7.3 Sexual Reproduction

Q 1 How is the process of pollination different from fertilisation?

✔ ANSWER

Pollination	Fertilisation
Transfer of pollen grains from the anther (stamen) to the stigma.	Fusion of the male germ-cell with the female gamete (egg) in the ovule.
Happens on the outside of the pistil, by wind, water or animals.	Happens inside the ovary, after the pollen tube reaches the ovule.
Comes first.	Follows pollination and forms the **zygote**.

Q 2 What is the role of the seminal vesicles and the prostate gland?

✔ ANSWER

They add their secretions along the path of the vas deferens, so the sperms are in a fluid. This fluid:

- makes the **transport of sperms easier**, and
- provides **nutrition** to the sperms.

Q 3 **What are the changes seen in girls at the time of puberty?**

✔ ANSWER

- Thick hair grows in the armpits and genital area; thinner hair on legs and arms.
- Breast size increases and the skin of the nipples darkens.
- Menstruation begins.
- Skin becomes oily and pimples may appear.

Q 4 **How does the embryo get nourishment inside the mother's body?**

✔ ANSWER

- The embryo gets nutrition from the mother's blood through a special tissue called the **placenta**, a disc embedded in the uterine wall.
- The placenta has villi on the embryo's side, surrounded by blood spaces on the mother's side.
- This gives a large surface area for glucose and oxygen to pass from mother to embryo; wastes pass from the embryo to the mother's blood.

Q 5 **If a woman is using a copper-T, will it help in protecting her from sexually transmitted diseases?**

✔ ANSWER

No. A copper-T is placed in the uterus only to prevent pregnancy.

It does not stop contact of body fluids, so it gives no protection against STDs such as gonorrhoea, syphilis or HIV-AIDS. A condom helps prevent them to some extent.

📝 Exercises

1 **Asexual reproduction takes place through budding in**
(a) Amoeba. (b) Yeast. (c) Plasmodium. (d) Leishmania.

✔ ANSWER

(b) Yeast. Yeast puts out small buds that separate and grow; Amoeba and Leishmania show binary fission, Plasmodium multiple fission.

2 **Which of the following is not a part of the female reproductive system in human beings?**

(a) Ovary (b) Uterus (c) Vas deferens (d) Fallopian tube

✔ ANSWER

(c) Vas deferens. It is part of the male reproductive system; it carries sperms from the testes.

3 **The anther contains**

(a) sepals. (b) ovules. (c) pistil. (d) pollen grains.

✔ ANSWER

(d) pollen grains. The anther of the stamen produces pollen grains.

4 **What are the advantages of sexual reproduction over asexual reproduction?**

✔ ANSWER

- DNA from two individuals combines, so it creates **more variation** and new combinations of characters.
- Variations help the species survive when the environment (niche) changes.
- Variations are the basis of **evolution** of new species.

- Harmful variations are less likely, since the combined variations come from living (healthy) individuals.

5 **What are the functions performed by the testis in human beings?**

✔ ANSWER

- **Formation of sperms** (male germ-cells).
- Secretion of the hormone **testosterone**, which regulates sperm formation and brings about the changes seen in boys at puberty.

6 **Why does menstruation occur?**

✔ ANSWER

- Every month the ovary releases one egg and the uterus lining becomes thick, spongy and rich in blood to receive a fertilised egg.
- If the egg is not fertilised (it lives about one day), this lining is not needed.
- So the lining slowly breaks down and comes out through the vagina as blood and mucous. This is menstruation; it occurs roughly every month and lasts about 2–8 days.

7 **Draw a labelled diagram of the longitudinal section of a flower.**

💡 SAMPLE ANSWER

Draw Fig. 7.7 of your book (longitudinal section of a flower). Labels to show:

- **Petal**, **Sepal**
- **Stamen**: **Anther** and **Filament**
- **Pistil (Carpel)**: **Stigma**, **Style**, **Ovary** with **Ovule(s)**
- Receptacle (base of the flower) (check the picture in your book)

8 **What are the different methods of contraception?**

✔ ANSWER

- **Mechanical barrier:** condom on the penis or similar covering in the vagina, so sperms do not reach the egg (also helps prevent STDs).
- **Hormonal (chemical):** oral pills change the hormonal balance so eggs are not released; may cause side-effects.
- **Intra-uterine devices:** loop or copper-T placed in the uterus; may irritate the uterus.
- **Surgical:** blocking the vas deferens in males (vasectomy) or the fallopian tube in females (tubectomy).

9 **How are the modes for reproduction different in unicellular and multicellular organisms?**

✔ ANSWER

- **Unicellular:** the single cell divides; e.g. binary fission (Amoeba, Leishmania), multiple fission (Plasmodium), budding (yeast). Simple cell division gives new individuals.
- **Multicellular:** cannot simply divide cell-by-cell because cells form tissues and organs. They use more complex methods: fragmentation (Spirogyra), regeneration and budding (Hydra, Planaria), vegetative propagation, spore formation (Rhizopus), and sexual reproduction with special reproductive organs and germ-cells.

10 **How does reproduction help in providing stability to populations of species?**

✔ ANSWER

- Reproduction produces new individuals that replace those that die, keeping up the population size.
- Consistent DNA copying keeps the body design that suits the organism to its niche.
- Small variations help some individuals survive if the niche changes, so the species is not wiped out.

11 **What could be the reasons for adopting contraceptive methods?**

✔ ANSWER

- To avoid unwanted pregnancy and to have gaps between children (protects the woman’s health).
- To keep the family small and control population growth for a better standard of living.
- To prevent sexually transmitted diseases (condoms).

🎉 **Chapter 7 done! Revise the answers once more tomorrow.**